

Latency Regression Postmortem

Checkout service, incident INC-1183

Reliability Engineering — 2026-05-28

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Summary

On 2026-05-26 the checkout service experienced elevated p99 latency (480 ms → 2.1 s) for 47 minutes, affecting an estimated 12% of purchase attempts. The root cause was an unbounded connection pool interacting with a slow downstream.¹

Timeline

Time (UTC)	Event
14:02	Deploy of checkout v3.8.0
14:09	p99 latency alert fires
14:18	On-call engaged; downstream identified as degraded
14:49	Connection pool cap lowered; latency recovers

Root cause

The release raised the database pool ceiling from 50 to 500. When the payments provider slowed, requests held connections while waiting, exhausting the database and creating a feedback loop.

¹Impact estimated from the difference between expected and observed completed-checkout counts during the window.

```
client → checkout → [pool: 500] → db
                        |
                    payments (slow) → held connections pile up
```

Corrective actions

- ☒ Cap the pool at 80 with a 250 ms acquire timeout
- ☒ Add a circuit breaker around the payments client
- ☐ Load-test the degraded-downstream scenario in staging
- ☐ Add a runbook entry for pool exhaustion

Lessons

Capacity limits are safety limits. A larger pool did not add headroom; it removed the backpressure that previously contained a downstream failure.